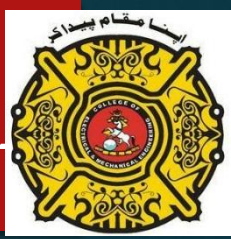




Professional Ethics



DR. MUHAMMAD ZEESHAN MIRZA

LECTURE - 6



Course Information



Course Name	Professional Ethics
Course Code	HU-222
Credit Hours	2
Email	zeeshanmirza21@hotmail.com
Visiting Hours	2-4 pm



Ethics in Engineering



- ▶ Where does ethics fit in engineering?





How Ethics Fits into Engineering?



Engineers

- ▶ **Build products** such as cell phones, home appliances, heart valves, bridges, & cars. In general, they advance society by building new technology
- ▶ **Develop processes**, such as the process to convert salt water into fresh water or the process to recycle bottles. These processes change how we live and what we can accomplish

Products and Processes have Consequences for Society



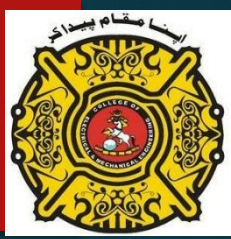
Decisions made by engineers usually have serious consequences to people often to multitudes of people

Ethics and ethical reasoning guide decision-making

If the process for refining gas produces too much toxins, it harms the local community



Japan Earthquake



- ▶ March 11, 2011 8.9 magnitude earthquake near Sendai, Japan
- ▶ The damage to the Fukushima I Nuclear Power Plant (*Fukushima Dai-ichi*) has led people worldwide to rethink the ethics of nuclear power.





Issue #1: Health and Safety



- ▶ Risks: Danger to current and future generations from leakage of radio-isotopes used in nuclear power
- ▶ Plutonium 239 is particularly toxic. Normally, 10 half lives (half life = 24,110 years) are required before a Pu 239 contaminated area is considered safe again, in the case of plutonium, roughly 250,000 years
- ▶ So, if Pu leaked, say, due to an earthquake it would cause a health risk for roughly 8000 generations



Issue #1: Health and Safety



- ▶ The possibility of medical science discovering a cure for cancer sometime in the current or next centuries studying radioactive isotopes
- ▶ The use of nuclear power may increase our knowledge of radioisotopes used for medical purposes (possible benefit?)



Issue #2: Depletion of Resources

- ▶ Fossil fuels, oil, natural gas and coal, are nonrenewable. These resources also affect the goal of health because of their impact on pollution and climate changes





Issue #3: Comparative Economic costs of Renewable Sources



- ▶ Renewable sources such as hydro electric power, wind power, solar power, geothermal heat, agricultural biomass and tides do not cause the environmental hazards that fossil fuels do
- ▶ But renewable sources must be balanced with the amount of energy needed to produce and maintain them and consequent environmental hazards



Typical Ethical Issues that Engineers Encounter



- ▶ Safety
- ▶ Acceptable risk
- ▶ Compliance
- ▶ Confidentiality
- ▶ Environmental health
- ▶ Conflict of interest
- ▶ Honesty/Dishonesty
- ▶ Societal impact



Two Dimensions of Ethics in Engineering



- ▶ Engineers need to be **socially responsible** when building products and processes for society
- ▶ Social responsibility requires **professional responsibility**
- ▶ *Engineering ethics is the study of the decisions, policies, and values that are morally desirable in engineering practice and research (Martin & Schinzinger)*



The Code of Ethics for Engineers



http://www.pec.org.pk/code_of_ethics.aspx

- ▶ There are many fields of engineering, such as Civil, Mechanical, Mechatronics, Electrical, Software, Industrial
- ▶ However, there are many ethical issues that arise across all of these fields of engineering.
- ▶ **The code of ethics for engineers pertains to engineers of all kinds**



Ethics Takes Practice



- ▶ Unlike robots, no one can just program you to be an ethical engineer that follows the codes
- ▶ It is possible to know the codes of ethics for engineering (or being a student), yet fail to follow them
- ▶ Ethical behavior is about practice and virtue. It is about going beyond the codes, and practicing behavior that leads to an ethical life



THE GULF COAST AND THE BP OIL SPILL

BUSINESS ETHICS CASE STUDY

A Case to Discuss



Important Information



- ▶ The Deepwater Horizon was a 10-year-old semi-submersible, mobile, floating, dynamically positioned drilling rig that could operate in waters up to 10,000 feet
- ▶ Built by South Korean company Hyundai Heavy Industries and owned by Transocean, and was chartered to BP from March 2008 to September 2013
- ▶ The well was situated in the Macondo Prospect, of the Gulf of Mexico, in the United States' exclusive economic zone
- ▶ BP was the operator and principal developer of the Macondo Prospect with a 65% share, while 25% was owned by Anadarko Petroleum Corporation, and 10% by MOEX Offshore 2007, a unit of Mitsui



The Explosion



- ▶ At approximately 9:45 pm CDT, on April 20, 2010, high-pressure methane gas from the well expanded into the drilling riser and rose into the drilling rig, where it ignited and exploded, engulfing the platform
- ▶ At the time, 126 crew members were on board: seven BP employees, 79 of Transocean, and employees of various other companies
- ▶ The explosion and subsequent fire resulted in the sinking of the Deepwater Horizon and the deaths of 11 workers; 17 others were injured



The Explosion





Causes



- ▶ The National Commission on the BP Deepwater Horizon Oil Spill and Offshore Drilling was established on 22 May to "consider the root causes of the disaster and offer options on safety and environmental precautions
- ▶ On 9 November 2010, a report by the Oil Spill Commission said that there had been "**a rush to completion**" on the well and criticized poor management decisions. "**There was not a culture of safety on that rig**" the co-chair said
- ▶ The panel found that BP, Halliburton, and Transocean had attempted to **work more cheaply** and thus helped to trigger the explosion and ensuing leakage



Causes



- ▶ “Whether purposeful or not, many of the decisions that BP, Halliburton, and Transocean made that increased the risk of the Macondo blowout clearly saved those companies **significant time (and money)**”
- ▶ Transocean, however, blamed BP for making the decisions before the actual explosion occurred and government officials for permitting those decisions
- ▶ Halliburton stated that it was acting only upon the orders of BP when it injected the cement into the wall of the well. It criticized BP for its failure to run a cement bond log test



Causes



- ▶ The US government report issued in September 2011 stated that **BP is ultimately responsible for the spill**, and that Halliburton and Transocean share some of the blame
- ▶ **Defective cement job**
- ▶ The investigation revealed numerous systems deficiencies, and acts and omissions by Transocean and its **Deepwater Horizon crew**, that had an adverse impact on the ability to prevent or limit the magnitude of the disaster
- ▶ Ignoring a pressure test that had failed

Causes

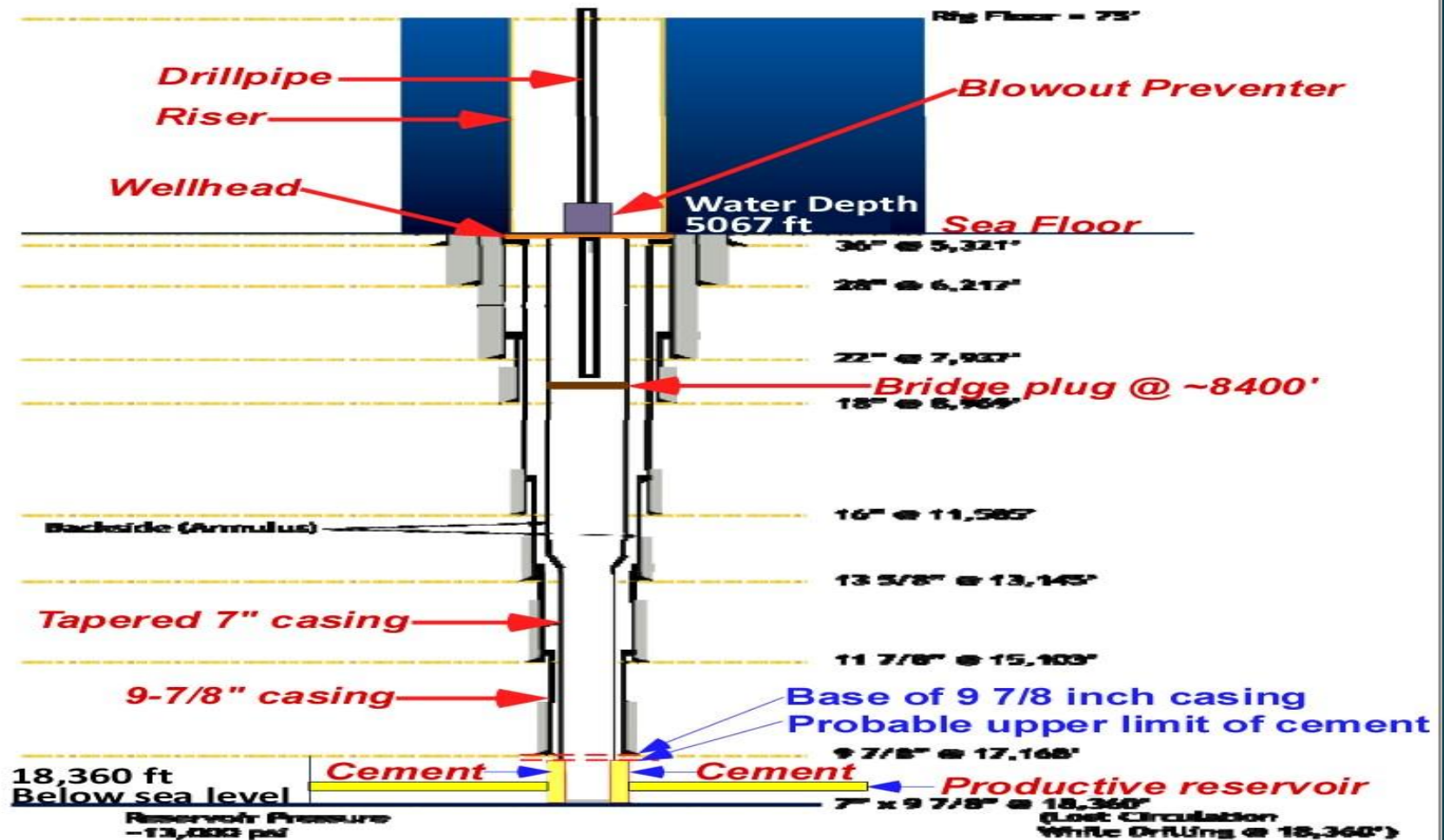


Figure 2. Mississippi Canyon Block 252 well schematic diagram (modified from www.tudorpickering.com/pdfs/tph.well.slides.pdf).



Environmental Impact



- ▶ A study in 2015 linked the sharp increase in dolphin deaths to the Deepwater Horizon oil spill
- ▶ On 12 April 2016, a research team reported that 88 percent of about 360 baby or stillborn dolphins within the spill area "had abnormal or under-developed lungs", compared to 15 percent in other areas





Health Consequences



- ▶ Studies have reported a variety of mental health issues, skin problems, breathing issues, coughing, and headaches
- ▶ A "significant percentage" of Gulf residents reporting mental health problems like anxiety, depression and PTSD
- ▶ A study that investigated the health effects among children in Louisiana and Florida living less than 10 miles from the coast found that more than a third of the parents reported physical or mental health symptoms among their children



Consequences on Economy



- ▶ Estimates of **lost tourism dollars** were projected to cost the Gulf coastal economy up to **22.7 billion** through 2013
- ▶ The Gulf of Mexico commercial fishing industry was estimated to have **lost \$247 million**
- ▶ During the crisis, BP gas stations in the United States reported a sales drop of between 10 and 40% due to backlash against the company

https://en.wikipedia.org/wiki/Deepwater_Horizon_oil_spill



Movie Adaptation



THANK YOU.....

